

Trigger resolution ϑ and model-update resolution q_r are independent

client i

server

clients

1 Local SGD

$$\delta_i^r = -\eta_r \sum_e g_{i,e}^r$$

2 Leaky state

$$\tilde{z}_i^r = \rho_r z_i^r + p_i \delta_i^r / \eta_r$$

3 Threshold + reset

$$|\tilde{z}_{ij}^r| \geq \vartheta \\ \Rightarrow c_{ij}^r = \pm 1, z_{ij}^{r+1} = 0$$

4 Server update

$$w^{r+1} = w^r + q_r \sum_i c_i^r$$

5 Exact sync

ordered sparse replay
or dense checkpoint

z_i^{r+1} persists

exact server model starts the next round